

Charts — matplotlib Bar Chart

Day 10

Lesson 10

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Course 02

Phase 2 — Data Lab

Time: 60–75 min

Mode: Self-study

Today's goals:

- ★ Install/import matplotlib
- ★ Make one bar chart from your command counts
- ★ Save figure as win_chart.png



Course 02 quest map: Day 10 of 18

Hook

Numbers convince parents and teammates. One clear chart beats ten vague guesses.

Learn



Learn

```
import matplotlib.pyplot as plt
labels = ["fight", "defend", "poison"]
values = [8, 3, 2]
plt.bar(labels, values)
plt.title("Commands used")
plt.savefig("win_chart.png")
Parent may run: pip install matplotlib once.
```



AI rules today

Predict/Run: AI off. **Investigate:** Socratic hints only. **Make:** you review every line.

Predict — Run — Investigate — Modify —> Make



Demo code

Folder: day10/

```
1 import matplotlib.pyplot as plt
2
3 labels = ["fight", "defend", "poison"]
4 values = [8, 3, 2]
5 plt.bar(labels, values, color=["#2E86AB", "#28A745", "#6F42C1"])
6 plt.ylabel("Times used")
7 plt.title("Commands in sample fights")
8 plt.tight_layout()
9 plt.savefig("win_chart.png")
10 print("Saved win_chart.png")
```

 Try it

In terminal, `cd course02/day10`, then predict output, then run: `python demos/demo_chart.py`

 Assignment

Work in: `day10/assignment/`

A — Predict What file appears after `savefig`?

B — Practice Run demo; open `win_chart.png`.

C — Level up Chart from YOUR fight data. Label axes. Save in `assignment/`.

D — Explain What story does your chart tell in one sentence?

 Checklist

- ☐ Re-read **Learn**; predict outputs (10 min)
- ☐ Assignments A–D (35–45 min)
- ☐ Run your program 3 times with different inputs
- ☐ Stuck 15 min → C-R-A-V (`../shared/prompts.md`)

Reflection

1. Easiest part today?
2. One line you can teach a friend.

Tomorrow: Change game balance using evidence.